Version Control

COMP1013 Analytics Programming
and
COMP7024 Programming for Data Science

Week 10 Practical

This lab requires the use of the software Git. Many modern computers come with Git installed; to check if it is installed, open a terminal (command prompt) and type `git`. If you don't have Git installed:

- Linux: sudo apt install git-all
- OSX: git –version (this will install the XCode command line tools (including git))
- Windows: Follow these instructions `https://git-scm.com/download/win`

1 *Creating a Git repository*

Git is used to manage version control for software projects (and can also be used for any other project, where files are updated over time). Projects are usually stored in their own directories, so Git will initialise within a directory, make all files in the directory and any in lower level directories, accessible to Git.

We are going to create a testing repository to test out how Git behaves.

1. Create a directory for the repository.
2. Initialise a Git repository within the directory.
3. Copy the epiSEIHCRD_combAge.csv file to the directory.
4. Add the file to the repository and commit the changes.
5. View the commit log to see your message.

2 *Adding a script to the repository*

1. Create an R script within this directory that reads the Sydney coronavirus data and plots the number of hospital beds needed between July and December of 2020.
2. Add the script to the repository and commit the changes.
3. View the log message.

3 Updating the script

1. Adjust the script so that the each month is shown in the x axis.
2. Commit the new changes (make sure to be providing informative log messages).
3. Change the plot line to be red and make sure that plot is labelled correctly.
4. Commit the new changes (make sure to be providing informative log messages).

4 Branching

The plot is exactly what is needed, but we want to examine if adding other curves will provide a better plot. We don't want to change the existing code, so we will make a branch and work on the new branch.

1. Create a branch in the repository and change to that branch.
2. Edit the script to add the number of critical and dead to the plot and commit the changes.
3. Change the y scale to be a square root scale. Make sure the plot looks presentable and then commit the changes.

5 Switching branches

You have been asked to update the plot so that the y axis has a log scale. To do this,

1. Switch back to the master branch, and make the changes.
2. Commit the changes.
3. Check the log.

6 Merging Branches

You have shown your work containing the plot the additional curves and the square root scale and it was well received. Place this work into the master branch by merging the branches.

1. Change to the master branch.
2. Merge the testing branch.
3. Delete the testing branch.
4. Tag this commit as version 1.

7 Sharing a repository

To share a repository, it must be placed where others can access it.

- It can we placed on a file server (with ssh access) or Web server (with https access).
- It can also be placed on a git Web server.

There are two widely used and free (no cost) to use remote git services.

- GitHub `https://github.com`
- BitBucket `https://bitbucket.org`

1. Set up an account at one of these (or both if you want).
2. Create a new repository using their Web interface and import your local git repository.
3. Share the repository URL with a classmate so they can clone it.

8 Connecting to the remote repository

By adding the Web hosted repository as a remote, we are now able to work on our own machine, and push changes to the remote server.

1. Check that the Web server is listed in the set of remotes.
2. Make a change to your code.
3. Commit the changes.
4. Push the changes to the remote.
5. View the Web interface and make sure that the changes have appeared.